Machine learning
Regression with Scikit-Learn

The aim of regression is to explain a variable Y by means of another variable X. For example, a person's salary can be explained by their level of higher education (although ...).
Linear regression is modeled by the following linear equation, which relates X and Y :

$$Y = \theta_0 + \theta_1 X + \epsilon$$

Where θ_0 and θ_1 are the model parameters and ϵ is the estimation error.

Since we have a single input variable, we're in the case of simple linear regression.
Multiple linear regression is when you have at least two input (or explanatory) variables.

The aim of this tutorial is to use the Scikit-learn module to perform simple linear regression.
In the following, we'll look at the modules provided by the Scikit-learn machine learning library for implementing regression functions in the case of simple linear regression.

Simple linear regression :

- Import scikit-learn (if not already installed, install it). Import the NumPy package and the LinearRegression class from scikit-learn.
- Using scikit-learn's make_regression, generate 100 samples, each with 3 features (note x) and 100 samples (note y) on which the model will train.
- Print the first 5 values of x and y
- Create an instance of the LinearRegression class
- Train it on our data set using the fit() method.
- Make some predictions with model_linReg.predict
- Calculate the accuracies.
- Then calculate the model's accuracy using the score() method.